

# Introduction to Robotics (Fall 2026)

ROB 345 / 549

## Assignment 1

Due Wed, Sept. 16th by midnight

Collaboration on the assignment is permitted (encouraged). But, you must turn in your own solutions for *all* portions of this assignment (theory and coding portions).

Additional HW policies are in the syllabus. In particular, no AI use for this assignment is permitted beyond the following: (1) questions about Python syntax, (2) explaining technical topics. Theory/programming assignments are meant to solidify your technical understanding. The drone-based assignments will provide opportunities to use generative AI tools such as coding agents.

### Part I: Theory

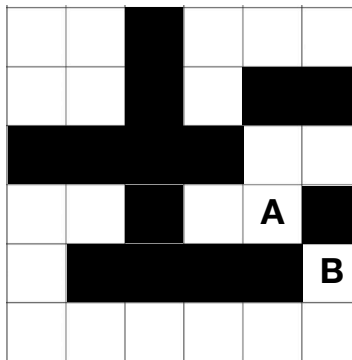


Figure 1: Environment for Problem 1.

**Problem 1.** Consider the environment shown in Figure 1. Our robot wants to get from cell A to cell B.

- (5 pts) Draw a graph (i.e., a set of vertices and edges connecting them) representing the environment shown in Figure 1. You can assume that the robot can only move up/down/left/right (i.e., the robot cannot move diagonally).
- (15 pts) Go through up to 5 iterations of Breadth First Search (BFS) on the graph you defined in part (a). For each iteration, write down the current vertex  $x$  being explored, its neighbors, and the vertices in  $Q$  (just as we did in class in Lecture 2). Explain the ordering you used to add vertices to the queue. If the algorithm terminates before you complete 5 iterations, you can just stop there.

(c) (15 pts) Now go through 5 iterations of Depth First Search (DFS) on the graph you defined in part (a). For each iteration, write down the current vertex  $x$  being explored, its neighbors, and the vertices in  $Q$ . You can stop if the algorithm terminates before you complete 5 iterations.

**Problem 2.** Consider the motion planning problem for the graph shown in Figure 2. There are five vertices, with the start vertex labeled “A” and the goal vertex labeled “B”. The cost of traversing each edge is also shown in the figure. You will use the A\* algorithm to find an optimal path for this graph. Recall that the A\* algorithm requires a cost-to-go estimate. Denote the cost-to-go estimate from vertex  $x$  by  $H(x)$ , and let:  $H(A) = 5$ ,  $H(x_1) = 3$ ,  $H(x_2) = 2$ ,  $H(x_3) = 4$ , and  $H(B) = 0$ .

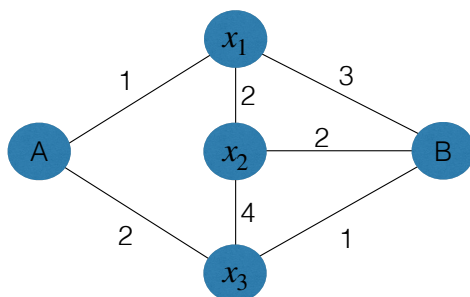


Figure 2: Graph for Problem 2.

(a) (8 pts) Walk through the steps of the A\* algorithm for this graph. You can do this by hand — you do not need to implement this in code. Run the algorithm until termination. For each iteration in the algorithm (see Lecture 2 slides for the full algorithm), write down the following: (i) the current vertex being explored (and an explanation for why this vertex was chosen from  $Q$ ), (ii) the neighbors of the current vertex being explored, (iii) the set of dead vertices, and (iv) the vertices in  $Q$ . At the end, write down the path that the algorithm found.

(b) (7 pts) Is the path that the A\* algorithm found optimal? If so, justify the claim. If it is not optimal, is there a way to fix the implementation in order to find an optimal path? In this case, either (1) find a fix and rerun the algorithm to successful completion, or (2) explain why a fix is not possible.

## Part II: Coding

Please follow the link to our GitHub page:  
<https://github.com/Princeton-Introduction-to-Robotics/F2026>  
 and follow the instructions in the “Install Instructions” and “Working on Assignments” sections. Please complete the problem in the “Lab1.ipynb” notebook. When you have completed this, please follow the submission instructions below.

## Submission Instructions

**Part I: Theory Submission** You can submit your written work for Part I (Theory) in Gradescope (accessed via Canvas) to “HW1: Theory”. You will have the option to submit each problem as

individual images or the entire assignment as one PDF. If submitting as one PDF, Gradescope will ask you to identify where problems are located. For more information, watch “For students: submitting PDF homework” from Gradescope’s Get Started page [https://www.gradescope.com/get\\_started](https://www.gradescope.com/get_started).

**Part II: Coding Submission** Please submit your completed Jupyter Notebook (Lab1.ipynb file) for Part II (Coding) in Gradescope to “HW1: Coding”.